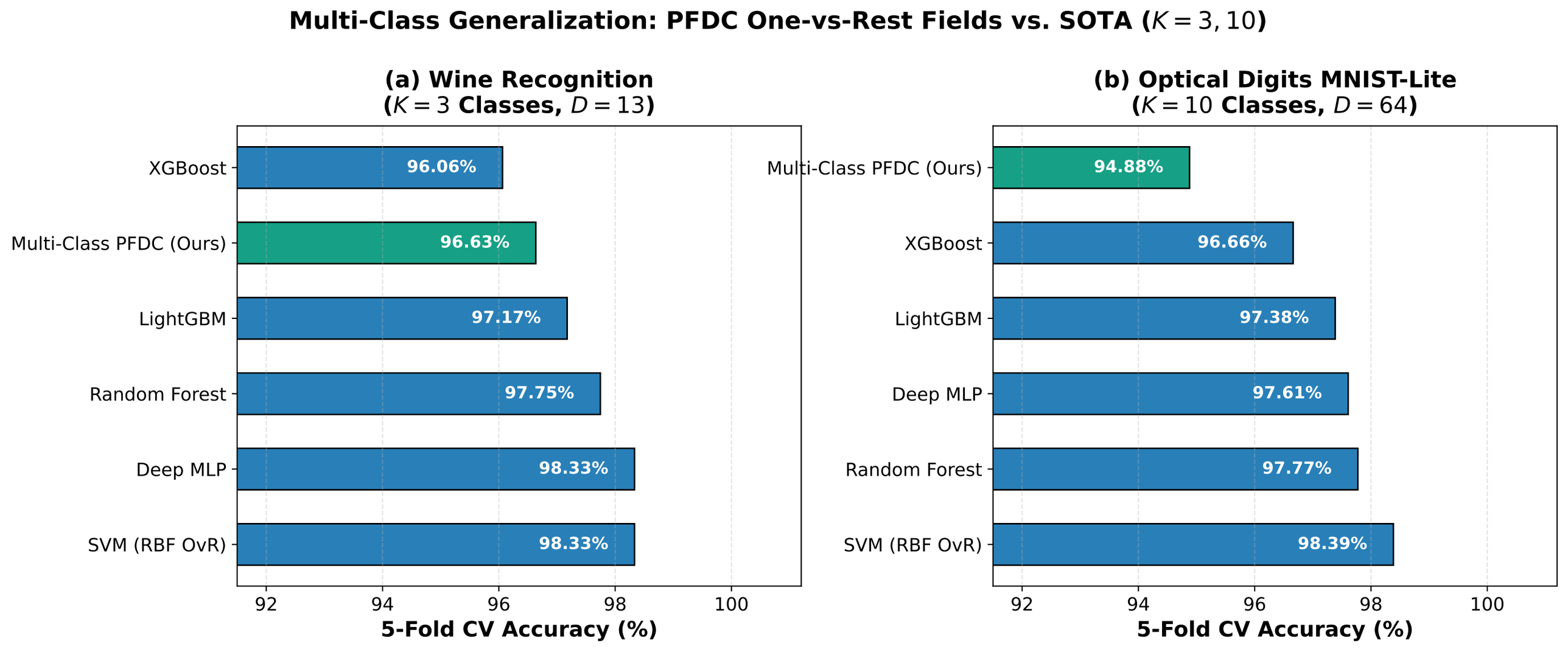
Multi-Class Generalization: PFDC One-vs-Rest Fields vs. SOTA ($K = 3, 10$)
(a) Wine Recognition
($K = 3$ Classes, $D = 13$)
XGBoost
Multi-Class PFDC (Ours)
LightGBM
Random Forest
Deep MLP
SVM (RBF OvR)
96.06%
96.63%
97.17%
97.75%
98.33%
98.33%
92
94
96
98
100
5-Fold CV Accuracy (%)
(b) Optical Digits MNIST-Lite
($K = 10$ Classes, $D = 64$)
Multi-Class PFDC (Ours)
XGBoost
LightGBM
Deep MLP
Random Forest
SVM (RBF OvR)
94.88%
96.66%
97.38%
97.61%
97.77%
98.39%
92
94
96
98
100
5-Fold CV Accuracy (%)